

A queue may be represented in the computer:

- **by means of linear arrays**
- **one-way lists.**

Queue representation using linear arrays:

- Queue will be maintained by a linear array QUEUE and
- two pointer variables:
 - FRONT → containing the location of the front element of the queue;
 - REAR → containing the location of the rear element of the queue.

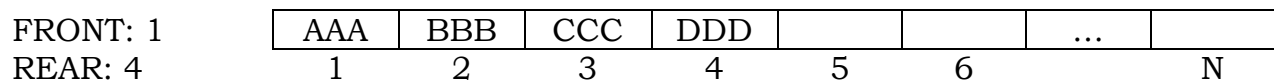


Fig (a) Array representation of Queue

- If we simply increment front and rear indices, then there may be problems, the front may reach the end of the array.
 - The solution to this problem is to increase front and rear in circular manner.

Inserting an item into the Queue:

QINSERT(QUEUE, N, FRONT, REAR, ITEM)

This procedure inserts an element ITEM into a queue.

1. [Queue already filled?]

If FRONT = 1 and REAR = N, or if FRONT = REAR + 1, then:

Write: OVERFLOW, and Return.

2. [Find new value of REAR.]

If FRONT := NULL, then : [Queue initially empty.]

Set FRONT := 1 and REAR :=1.

Else if REAR = N, then:

Set REAR :=1.

Else:

Set REAR := REAR + 1.

[End of If structure.]

3. Set QUEUE[REAR] := ITEM. [This inserts new element]
4. Return.

Removing an item from the queue:

QDELETE(QUEUE, N, FRONT, REAR, ITEM)

This procedure deletes an element from a queue.

1. [Queue already empty?]
If FRONT := NULL, then: Write: UNDERFLOW, and Return.
2. Set ITEM:= QUEUE[FRONT].
3. [Find new value of FRONT.]
If FRONT = REAR, then: [Queue has only one element to start.]
Set FRONT:= NULL and REAR:= NULL
Else if FRONT= N, then:
Set FRONT := 1.
Else:
Set FRONT:= FRONT + 1
[End of If Structure]
4. Return.

Queue Variation:

- **DEQUES:**
 - A deque (double ended queue) is a linear list in which elements can be added or removed at either end but not in the middle.
- **Priority Queues:**
 - A priority queue is a collection of elements such that each element has been assigned a priority and such that the order in which elements are deleted and processed comes from the following rules:
 - An element of higher priority is processed before any element of lower priority.
 - Two elements with the same priority are processed according to the order in which they were added to the queue.
 - A prototype of a priority queue is a timesharing system: programs of high priority are processed first and programs with the same priority form a standard queue.